Streamlining consortium-based analyses of metabolomics data with COMETS Analytics

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Metabolomics Epidemiology Requires Analytics Infrastructure

- *Metabolomics data generated in thousands of patients*
- *Important to look for consistent biomarkers or risk factors across multiple large studies*

Issues and Needs

Security

Many institutes cannot transfer data

Expertise

Working with original data requires extensive expertise

Robustness

Large amounts of data with varying study designs

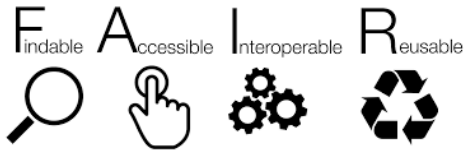
Rigor

Accuracy and traceability of results

Solution: Federated analyses model that eliminates data transfer and pools results centrally

COMETS Analytics

COMETS Analytics Design Principles



Minimize burden on analyst to harmonize data and run models (without errors)

- ❑ Standardized formats for data, models, results
- ❑ Extensive and automated warning/error checking on data and models
- ❑ Control of definitions for data and models

Provide a robust and reproducible platform

- ❑ Consistent results formats and traceable calculations
- ❑ Thorough testing
- ❑ Contact email: comets.analytics@gmail.com

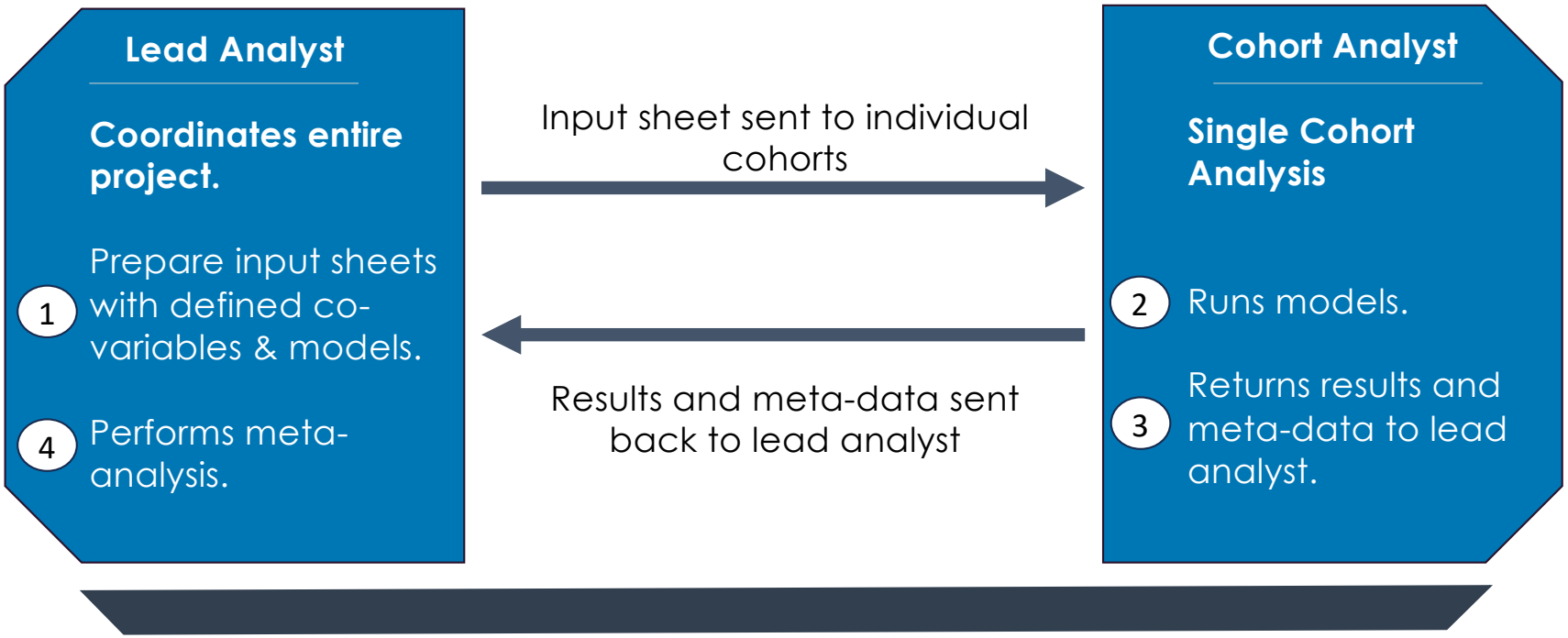
Ensure software is usable and accessible

- ❑ Public, user-friendly access through a local installation of the R package and web interface
- ❑ Extensive documentation with examples (R functions, vignette, tutorial)

Ensure privacy of data

- ❑ R package can be run locally on private servers
- ❑ Web interface provides secure upload of data (Amazon S3 server 256-bit advanced encryption standard) with results deleted after 7 days.

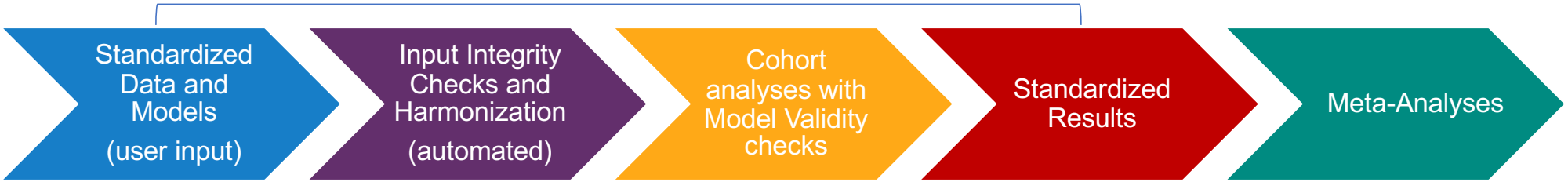
Roles and for data analysts



COMETS Data Infrastructure and Early Career Interest Groups support data harmonization, bug fixes, troubleshooting

Implementation of Design Principles

Same process and input template for each cohort



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Author Notes

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Input File (Excel) Standardizes Data and Models

Information on Metabolites	Information on Participants	Information on Models
Metabolites Metabolite information	**Subjects** Subject information	**Models** Model specifications
Subject Metabolites Metabolite levels per subject	**VarMap** Definitions for variable naming and coding	**Model Types** Advanced options for specifying models

Basics of Input File

	A	B	C	D	E	F	G
1	metabid	metabolite_name	SUPER_PATHWAY	SUB_PATHWAY	COMP_ID	PLATFORM	HMDB_ID
2	ACETOACETATE	acetoacetate	Lipid	Ketone Bodies	33963	GC/MS	HMDB00060
3	ACETYLCARNITINE	acetylcarnitine	Lipid	Fatty Acid Metabolism(Acyl Carnitine)	32198	LC/MS Pos	HMDB00201
4	ACISOGA	acisoga	Amino Acid	Polyamine Metabolism	43258	LC/MS Pos	NA
5	ADENINE	adenine	Nucleotide	Purine Metabolism, Adenine containing	554	LC/MS Pos	HMDB00034
6	ADENOSINE	adenosine	Nucleotide	Purine Metabolism, Adenine containing	555	LC/MS Pos	HMDB00050
7	ADRENATE__22_4N6_	adrenate (22:4n6)	Lipid	Polyunsaturated Fatty Acid (n3 and n6)	32980	LC/MS Neg	HMDB02226
8	ADSGEGDFXAEGGGVR	ADSGEGDFXAEGGGVR*	Peptide	Fibrinogen Cleavage Peptide	33084	LC/MS Pos	NA
9	ALANINE	alanine	Amino Acid	Alanine and Aspartate Metabolism	1126	LC/MS Pos	HMDB00161
10	ALANYLLEUCINE	alanylleucine	Peptide	Dipeptide	37093	LC/MS Pos	HMDB28691
11	ALLANTOIN	allantoin	Nucleotide	Purine Metabolism, (Hypo)Xanthine/Inosine containing	1107	LC/MS Polar	HMDB00462
12	ALLIIN	alliin	Xenobiotics	Food Component/Plant	41494	LC/MS Pos	HMDB33592
13	ALLO_ISOLEUCINE	allo-isoleucine	Amino Acid	Leucine, Isoleucine and Valine Metabolism	47034	GC/MS	NA
14	ALLO_THREONINE	allo-threonine	Amino Acid	Glycine, Serine and Threonine Metabolism	15142	LC/MS Polar	HMDB04041
15	ALPHA_CEHC	alpha-CEHC	Cofactors and Vitamins	Tocopherol Metabolism	48188	LC/MS Neg	HMDB01518
16	ALPHA_CEHC_GLUCURONIDE	alpha-CEHC glucuronide*	Cofactors and Vitamins	Tocopherol Metabolism	39346	LC/MS Neg	NA
17	ALPHA_CEHC_SULFATE	alpha-CEHC sulfate	Cofactors and Vitamins	Tocopherol Metabolism	47666	LC/MS Neg	NA
18	ALPHA_HYDROXYCAPROATE	alpha-hydroxycaproate	Lipid	Fatty Acid, Monohydroxy	37073	LC/MS Neg	HMDB01624
19	ALPHA_HYDROXYISOCAPROATE	alpha-hydroxyisocaproate	Amino Acid	Leucine, Isoleucine and Valine Metabolism	22132	LC/MS Neg	HMDB00746
20	ALPHA_HYDROXYISOVALERATE	alpha-hydroxyisovalerate	Amino Acid	Leucine, Isoleucine and Valine Metabolism	33937	LC/MS Neg	HMDB00407
21	ALPHA_KETOBUTYRATE	alpha-ketobutyrate	Amino Acid	Methionine, Cysteine, SAM and Taurine Metabolism	4968	LC/MS Polar	HMDB00005
22	ALPHA_KETOGLUTARATE	alpha-ketoglutarate	Energy	TCA Cycle	528	LC/MS Polar	HMDB00208
23	ALPHA_TOCOPHEROL	alpha-tocopherol	Cofactors and Vitamins	Tocopherol Metabolism	1561	GC/MS	HMDB01893
24	ANDROSTERONE_SULFATE	androsterone sulfate	Lipid	Steroid	31591	LC/MS Neg	HMDB02759
25	ANDRO_STEROID_MONOSULFATE_1	andro steroid monosulfate (1)*	Lipid	Steroid	32827	LC/MS Neg	HMDB02759
26	ARABINOSE	arabinose	Carbohydrate	Pentose Metabolism	575	GC/MS	HMDB00646
27	ARABITOL	arabitol	Carbohydrate	Pentose Metabolism	38075	GC/MS	HMDB01851
28	ARABONATE	arabonate	Cofactors and Vitamins	Ascorbate and Aldarate Metabolism	37516	GC/MS	HMDB00539
29	ARACHIDATE__20_0_	arachidate (20:0)	Lipid	Long Chain Fatty Acid	1118	LC/MS Neg	HMDB02212

Metabolites | SubjectMetabolites | SubjectData | VarMap | Models | Model_Types | +

COMETS Analytics

Data Harmonization - Covariables

Variables must be defined consistently across cohorts for meta-analyses

	A	B	C	D	E	F
1	**VARREFERENCE**	**VARDEFINITION**	**VARTYPE**	**ACCEPTED_VALUES**	**COHORTVARIABLE**	**COHORTNOTES**
2	id	cohort subject id	NA		SAMPLE_ID	Sample Identifier
3	metabolite_id	metabolite identifier / variable name	NA		metabid	column header in metabolites sheet
4	age	Age at Entry (In Years)	continuous	(0, Inf)	age	No missing data allowed
5	female	Female (0=male, 1=female)	categorical	0, 1	sex	No missing data allowed
6	smk_grp	Smoking status (0=never smoker, 1=former smoker, 2=current smoker, 3=missing)	categorical	0, 1, 2, 3	smoking	Only missing values encoded as "" will be counted as missing when the data is read in; this also applies to all variables below with a non-blank missing value.

Lead analyst provide definitions of covariables

Cohort analyst provides internal names and notes

Rigor and reproducibility – Input sheets provide a guidance/dictionary of covariable definitions

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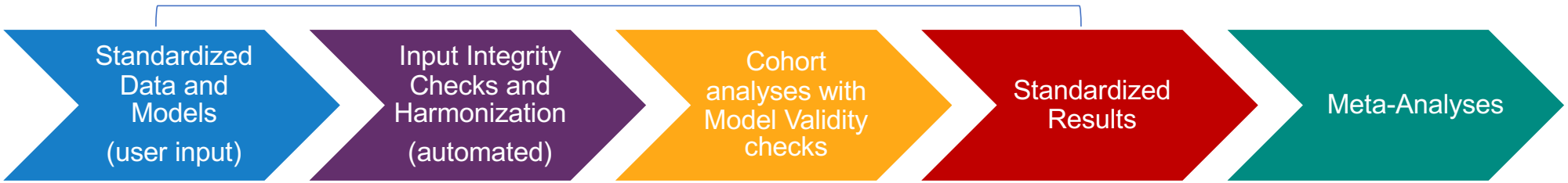
Implementation of Design Principles

Same process and input template for each cohort



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Integrity Checks on Data and Models

Goal: Ensure that data input is sufficient for analytical integrity

Errors - process is halted

- Software cannot find appropriate variables (incorrect Variable Mapping sheet)
- Metabolites or samples have missing meta data
- Models defined are not correct (e.g. adjustment variable is also an outcome variable)

Warnings - process continues

- Variables are appropriately dropped in models (e.g. no variance, etc.)
- Models are appropriately dropped (e.g. low sample size, etc.)
- Model parameters related to filtering are left blank

> 100 integrity checks performed automatically in a few seconds with meaningful messages!

COMETS Analytics

Data Harmonization is required for meta-analyses

✧ Analysis will proceed whether or not covariables and metabolite names are harmonized!

VARREFERENCE

BMI (kg/m^2)

COHORT VARIABLES

Body Mass Index (kg/m^2)

BMI

bmi_metric

BMI (lb/in^2)

Individual_BMI

	A	B	C	D	E	F
1	VARREFERENCE	VARDEFINITION	VARTYPE	ACCEPTED_VALUES	COHORTVARIABLE	COHORTNOTES
2	id	cohort subject id	NA		SAMPLE_ID	Sample Identifier
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5	female	Female (0=male, 1=female)	categorical	0, 1	sex	No missing data allowed
6	smk_grp	Smoking status (0=never smoker, 1=former smoker, 2=current smoker, 3=missing)	categorical	0, 1, 2, 3	smoking	Only missing values encoded as "" will be counted as missing when the data is read in; this also applies to all variables below with a non-blank missing value.

Harmonized Name

D-Glucose

Alternate names provided by cohorts

15903

D-Glucose

C00221

D Glucose

HMDB0000122

Glucose

Curated Harmonization

Automated Harmonization

COMETS Analytics

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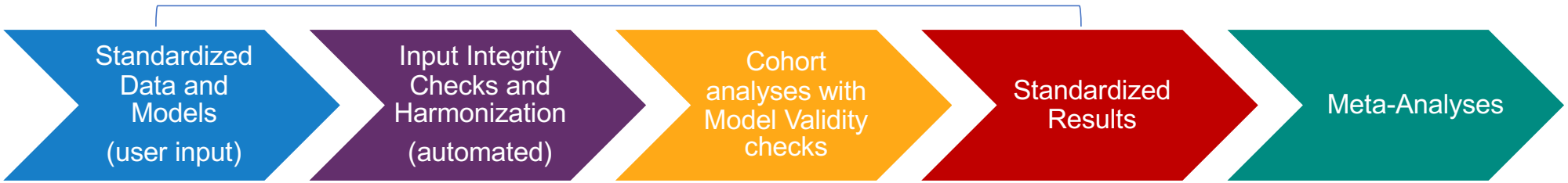

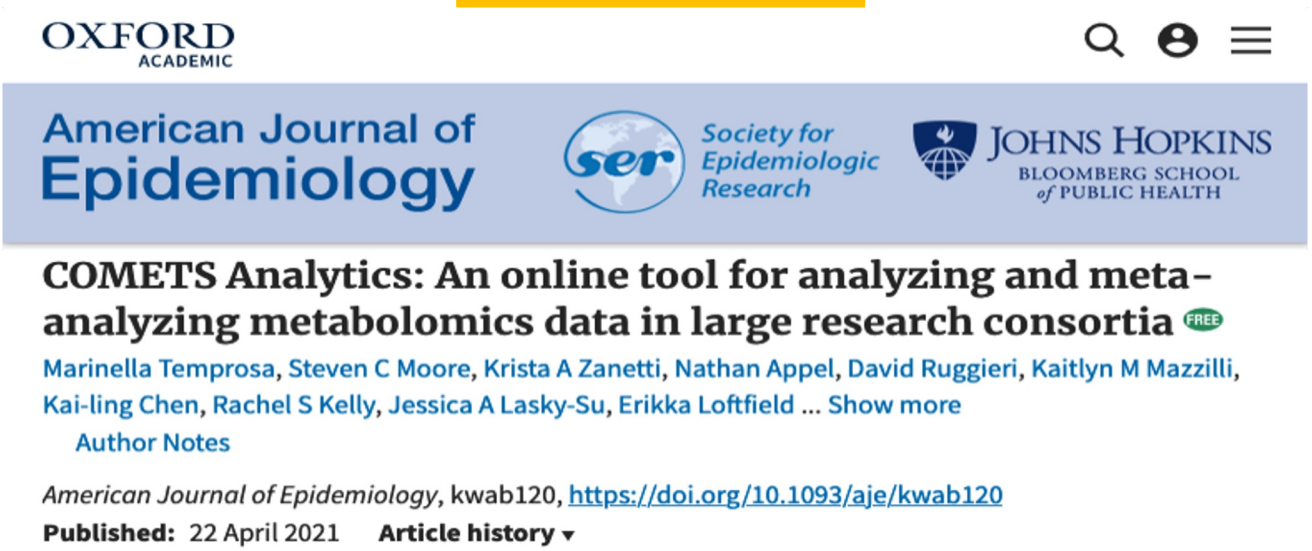

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Input sheet standardizes models run

A	B	C	D	E	F	G	H	I	J
MODEL	**OUTCOMES**	**EXPOSURE**	**EXPOSURE_ REFERENCE**	**ADJUSTMENT**	**STRATIFICATION**	**WHERE**	**TIME**	**GROUP**	**MODEL_TYPE**
1 Age	All metabolites	age							**corr (spearman)**
2 Age Spearman Min 10 Subjects	All metabolites	age							**corr (user-defined)**
3 Age Multivariable adjusted	All metabolites	age		smk_grp bmi_grp race_grp educ_grp alc_grp horm_curr					**corr (spearman)**
4 Age Multivariable adjusted stratified	All metabolites	age		smk_grp race_grp educ_grp alc_grp horm_curr	bmi_grp				**corr (spearman)**
5 Age Multivariable adjusted stratified subset	All metabolites	age		female smk_grp bmi_grp race_grp educ_grp alc_grp multivitamin fasted	horm_curr	age < 70			**corr (spearman)**
6 All pairwise metabolites	All metabolites	All metabolites							**corr (spearman)**
7 Poisson regression	n_visits	All metabolites		age					**GLM poisson**
8 Logistic regression	nested_case	All metabolites		age					**GLM binomial**
9 Survival model	event	All metabolites		age			time		**survival (efron)**
10 Conditional logistic	nested_case	All metabolites		age				matchedSet	**cond logistic (efron)**

* Names used in the model sheet are consistent across all cohorts and integrity checks enable running of models without errors

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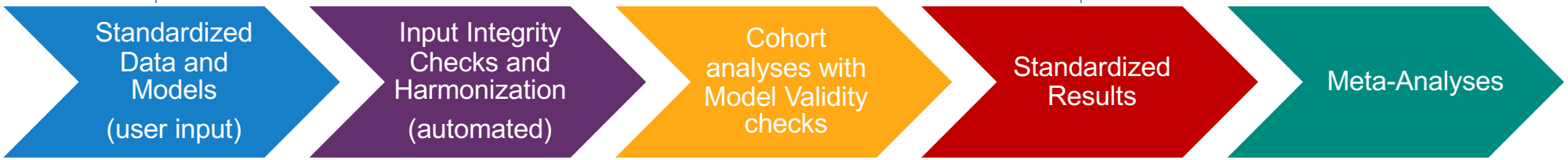
Implementation of Design Principles

Same process and input template for each cohort



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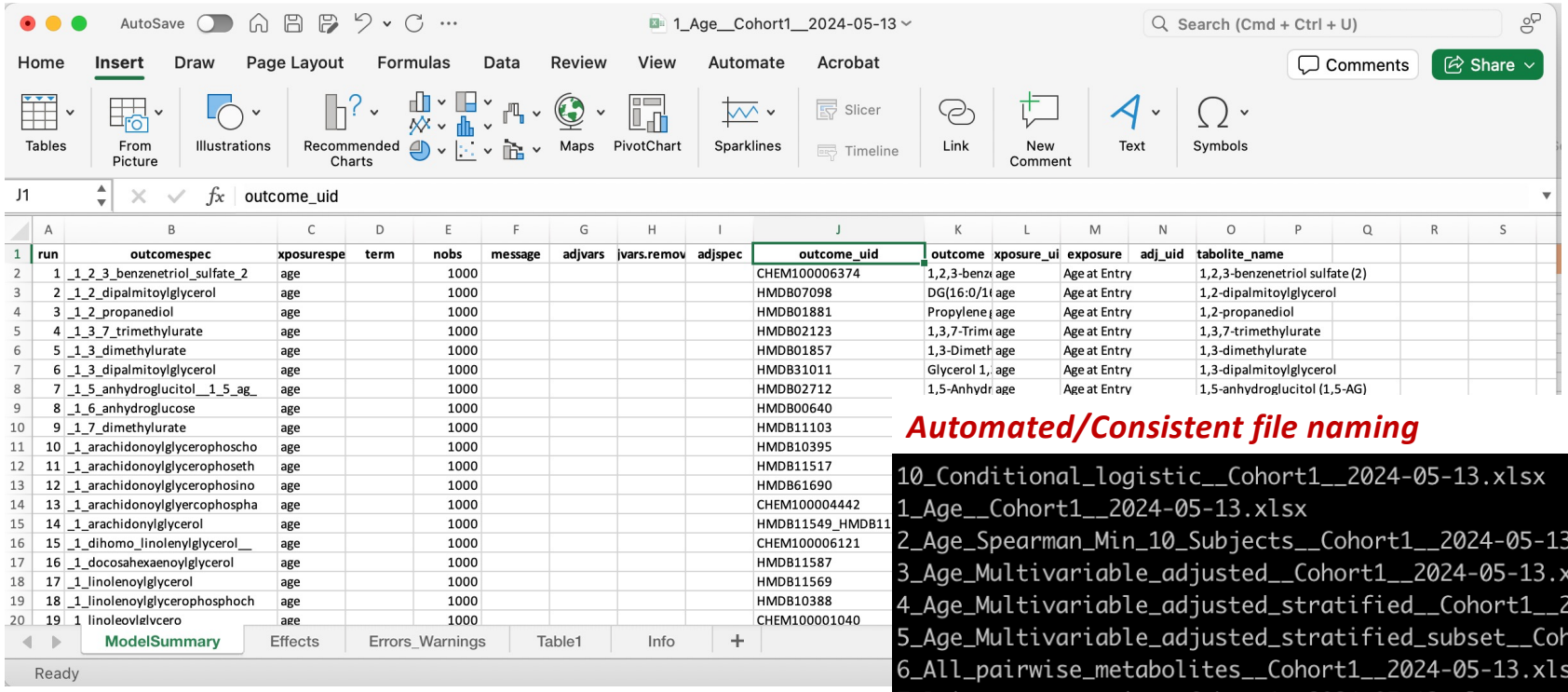
Standardized Model Outputs

Model Summary	• Outcome, exposure, and adjustment variables • # observations
Effects	• Estimates (e.g., odds ratios) • Statistics (e.g., confidence intervals) and p-values
Errors & Warnings	• Report on errors and warnings for tracking • e.g., variables dropped due to insufficient numbers
Table 1	• Demographic table with number of observations, min/max values, medians, etc.
Other Information	• metadata related to the software settings • e.g., version, cohort name, parameters used, etc.

Standardized Results Outputs



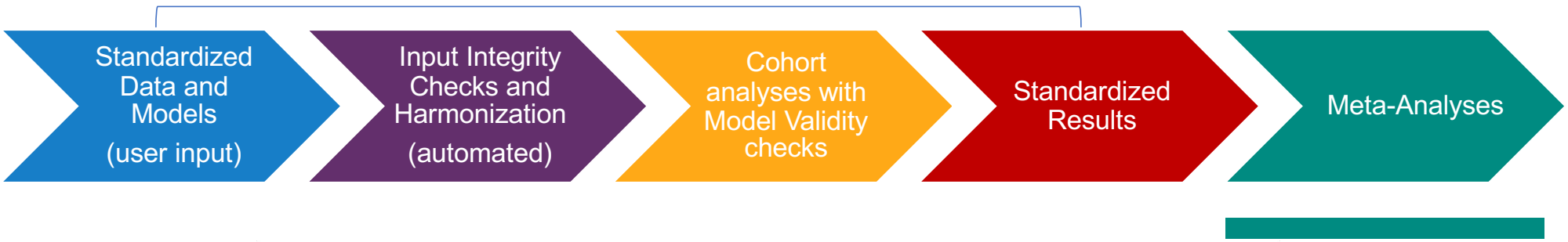
run	outcomespec	xposurespe	term	nobs	message	adjvars	jvars.remov	adjspec	outcome_uid	outcome	xposure_ui	exposure	adj_uid	tabolite_name
1	_1_2_3_benzenetriol_sulfate_2	age		1000					CHEM100006374	1,2,3-benz	age	Age at Entry		1,2,3-benzenetriol sulfate (2)
2	_1_2_dipalmitoylglycerol	age		1000					HMDB07098	DG(16:0/1	age	Age at Entry		1,2-dipalmitoylglycerol
3	_1_2_propanediol	age		1000					HMDB01881	Propylene	age	Age at Entry		1,2-propanediol
4	_1_3_7_trimethylurate	age		1000					HMDB02123	1,3,7-Trim	age	Age at Entry		1,3,7-trimethylurate
5	_1_3_dimethylurate	age		1000					HMDB01857	1,3-Dimeth	age	Age at Entry		1,3-dimethylurate
6	_1_3_dipalmitoylglycerol	age		1000					HMDB31011	Glycerol 1,	age	Age at Entry		1,3-dipalmitoylglycerol
7	_1_5_anhydroglucitol__1_5_ag_	age		1000					HMDB02712	1,5-Anhydr	age	Age at Entry		1,5-anhydroglucitol (1,5-AG)
8	_1_6_anhydroglucose	age		1000					HMDB00640					
9	_1_7_dimethylurate	age		1000					HMDB11103					
10	_1_arachidonoylglycerophoscho	age		1000					HMDB10395					
11	_1_arachidonoylglycerophoseth	age		1000					HMDB11517					
12	_1_arachidonoylglycerophosino	age		1000					HMDB61690					
13	_1_arachidonoylglyercophospha	age		1000					CHEM100004442					
14	_1_arachidonylglycerol	age		1000					HMDB11549_HMDB11					
15	_1_dihomo_linolenylglycerol__	age		1000					CHEM100006121					
16	_1_docosahexaenoylglycerol	age		1000					HMDB11587					
17	_1_linolenoylglycerol	age		1000					HMDB11569					
18	_1_linolenoylglycerophosphoch	age		1000					HMDB10388					
19	_1_linoleoylglycero	age		1000					CHEM100001040					

Automated/Consistent file naming

```
10_Conditional_logistic__Cohort1__2024-05-13.xlsx
1_Age__Cohort1__2024-05-13.xlsx
2_Age_Spearman_Min_10_Subjects__Cohort1__2024-05-13.xlsx
3_Age_Multivariable_adjusted__Cohort1__2024-05-13.xlsx
4_Age_Multivariable_adjusted_stratified__Cohort1__2024-05-13.xlsx
5_Age_Multivariable_adjusted_stratified_subset__Cohort1__2024-05-13.xlsx
6_All_pairwise_metabolites__Cohort1__2024-05-13.xlsx
7_Poisson_regression__Cohort1__2024-05-13.xlsx
8_Logistic_regression__Cohort1__2024-05-13.xlsx
9_Survival_model__Cohort1__2024-05-13.xlsx
```

Implementation of Design Principles

Same process and input template for each cohort



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COMETS Analytics Facilitates Meta-Analyses

Key Design Attributes

- Standardized definitions of covariables and models (by lead analyst).
- Automated integrity checks and metabolite name harmonization.
- Format of outputs is identical for each individual cohort analyses.
- Naming of output files is consistent for each individual cohort analyses.

Simple Steps for Analyses

1. Load Data - readCOMETSinput()
2. Define model – getModelData()
   - Interactively by specifying exposures and other criteria
   - By referring to a model pre-defined in the input Excel sheet
3. Run models – runModel() for reach cohorts
4. Run meta-analysis – runMeta()

COMETS Analytics

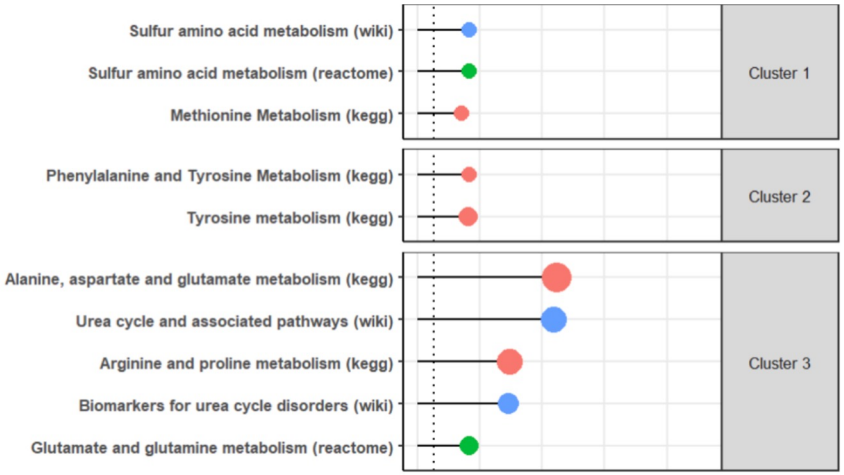
Enrichment Analyses for Data Interpretation

Annotations for >230K metabolites, ~16K genes/enzymes, >53K pathways
> 63K reactions, > 217K chemical property records

https:rampdb.nih.gov

RaMP Vignette
https://bit.ly/3UA65EA

Chemical and Biochemical Pathway Enrichment embedded into COMETS Analytics



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NCI DCCPS
NCI DCEG
NIH STRIDES

Users and other COMETS Interest Groups!
Thank you for your feedback!

Contact us at comets.analytics@gmail.com

COMETS Analytics

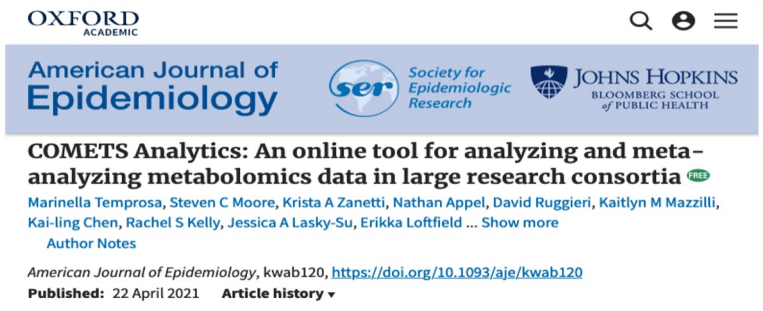
Break – see you back at 11 am EST!

COMETS Analytics Public and Secure Web Interface

https://bit.ly/3wtd0aQ

Example Code and Documents

https://bit.ly/NIH_NCATS_Met_Epi_workshop



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COMETS Analytics R Package and Documentation

https://bit.ly/4biAXAz